# LIMITING REAGENTS 2

1 What mass of calcium hydroxide is formed when 10.0 g of calcium oxide reacts with 10.0 g of water?

$CaO + H_2O \rightarrow Ca(OH)_2$

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2 What mass of magnesium bromide is formed when 1.00 g of magnesium reacts with 5.00 g of bromine?

$Mg + Br_2 \rightarrow MgBr_2$

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3 What mass of copper is formed when 2.00 g of copper(II) oxide reacts with 1.00 g of hydrogen?

$CuO + H_2 \rightarrow Cu + H_2O$

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**4** What mass of sodium fluoride is formed when 2.30 g of sodium reacts with 2.85 g of fluorine? $2Na + F_2 \rightarrow 2NaF$

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**5** What mass of iron is formed when 8.00 g of iron(III) oxide reacts with 2.16 g of aluminium? $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$

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**6** What mass of aluminium chloride is formed when 13.5 g of aluminium reacts with 42.6 g of chlorine? $2Al + 3Cl_2 \rightarrow 2AlCl_3$

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| Area | Strength | To develop | Area | Strength | To develop | Area | Strength | To develop |
|---|---|---|---|---|---|---|---|---|
| Done with care and thoroughness | | | Can find moles from mass | | | Can find mass of product | | |
| Shows suitable working | | | Can identify limiting reagent | | | Gives units | | |
| Can work out $M_r$ | | | Can deduce moles of product | | | | | |